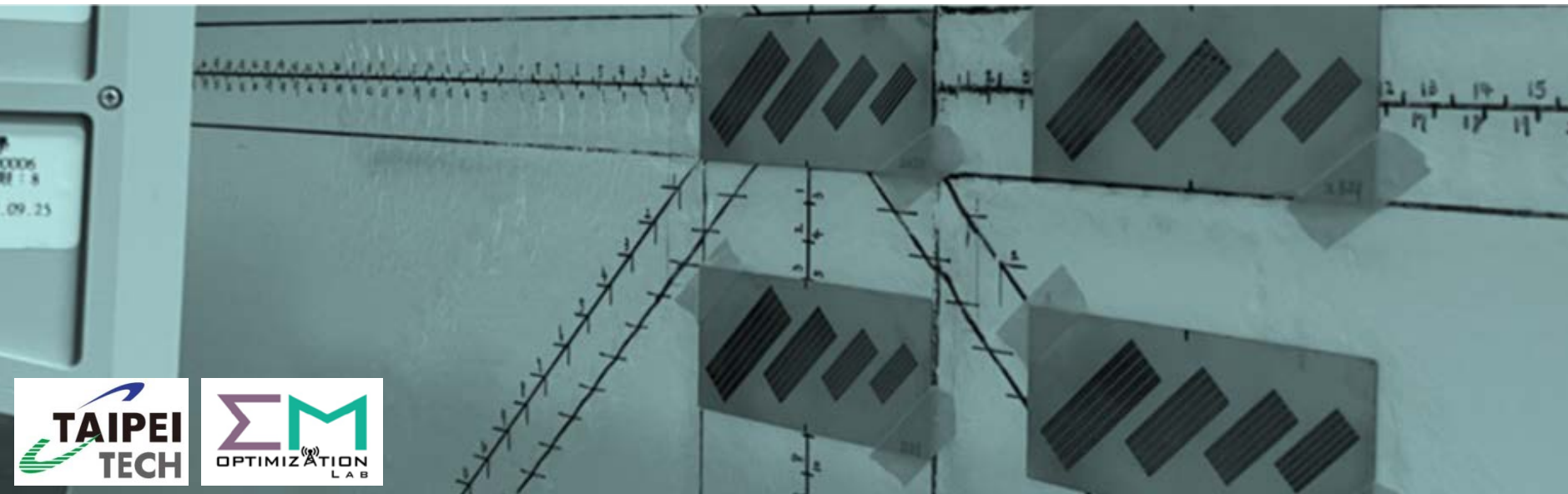
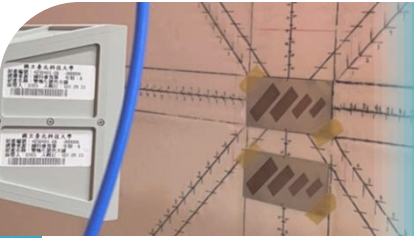


Linear Algebra

線性代數



Chapter 6: Singular Value Decomposition
Yen-Sheng Chen (陳晏笙)
OCW Course Content



Agenda

1

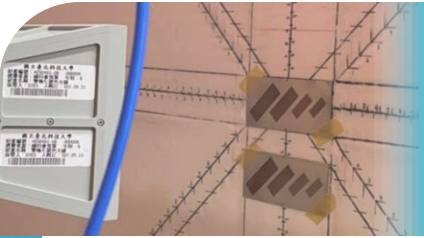
Concept

2

Procedure

3

Applications



Agenda

1

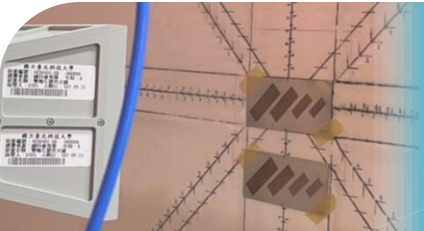
Concept

2

Procedure

3

Applications



Limitations in Previous Chapters

Chapter 1

What if the inverse of a matrix $\mathbf{A}^{-1}$ fails to exist?

Chapter 2

How do we find bases for 4 subspaces numerically?

Chapter 3

$$\mathbf{A}^T \mathbf{A} \hat{\mathbf{x}} = \mathbf{A}^T \mathbf{b}$$

What if $\mathbf{A}$ has dependent columns?

Chapter 5

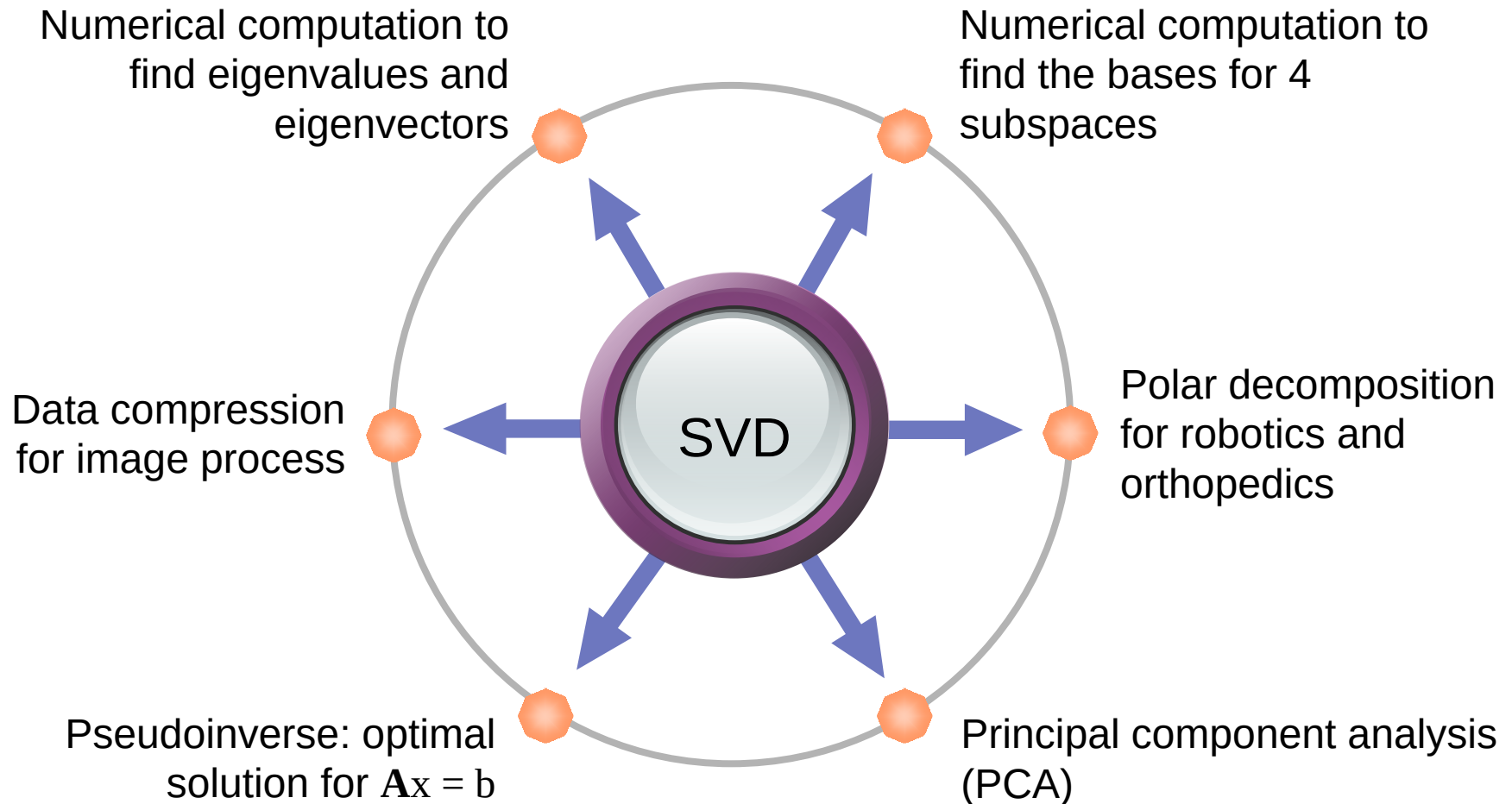
How do we define λ for rectangular $\mathbf{A}$?

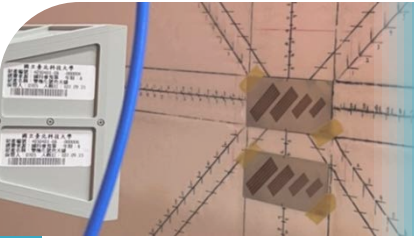
Chapter 5

Can we factorize ANY $\mathbf{A}$ as $\mathbf{Q}_1 \mathbf{D} \mathbf{Q}_2^T$?

SVD

Applications of SVD





Agenda

1

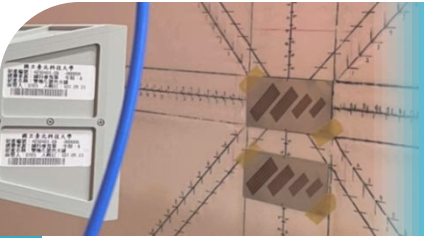
Concept

2

Procedure

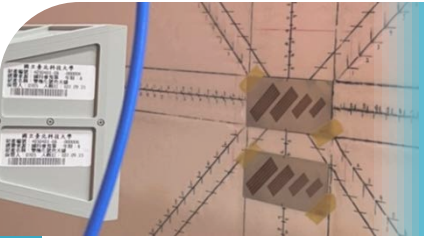
3

Applications



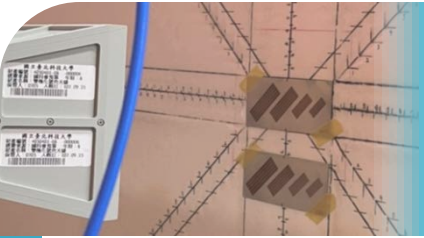
Basics of SVD: $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}\mathbf{A}^T$

- Recall Ch. 3: Characteristics about $\mathbf{A}^T\mathbf{A}$
 - ◆ It is an important matrix in normal equation: $\mathbf{A}^T\mathbf{A}\hat{\mathbf{x}} = \mathbf{A}^T\mathbf{b}$
 - ◆ Symmetric
 - ◆ $\mathbf{A}^T\mathbf{A}$ has the same nullspace as $\mathbf{A}$
 - ◆ If $\mathbf{A}$ has linearly independent columns, $\mathbf{A}^T\mathbf{A}$ is square and invertible
 - ◆ $\hat{\mathbf{x}}$ can be solved when $\mathbf{A}$ has full column rank (n independent columns)
- Let's generalize these ideas and include another important matrix: $\mathbf{A}\mathbf{A}^T$



Basics of SVD: $A^T A$ and AA^T

- $A^T A$:
 - ◆ Symmetric and positive semidefinite
 - ◆ Invertible positive definite when A has independent columns
 - ◆ $A^T A$ has the same nullspace as A
 - ◆ Factorization of $A^T A$: $A^T A = Q_2 \Lambda_2 Q_2^T$
- AA^T :
 - ◆ Symmetric and positive semidefinite
 - ◆ Invertible positive definite when A has independent rows
 - ◆ AA^T has the same left nullspace as A (that is, the nullspace of A^T)
 - ◆ Factorization of AA^T : $AA^T = Q_1 \Lambda_1 Q_1^T$



Basics of SVD: $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}\mathbf{A}^T$

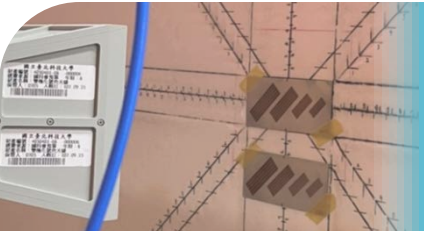
- $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}\mathbf{A}^T$ share the same **nonzero** eigenvalues
- For the same eigenvalue, the eigenvector $\mathbf{x}$ for $\mathbf{A}^T\mathbf{A}$ corresponds to the eigenvector $\mathbf{A}\mathbf{x}$ for $\mathbf{A}\mathbf{A}^T$
- Diagonalization for the 2 matrices:

$$\mathbf{Q}_2^T (\mathbf{A}^T \mathbf{A}) \mathbf{Q}_2 = \mathbf{\Lambda}_2, \quad \mathbf{Q}_1^T (\mathbf{A} \mathbf{A}^T) \mathbf{Q}_1 = \mathbf{\Lambda}_1$$

$$\mathbf{\Lambda}_2 = \begin{bmatrix} \lambda_1 & & & & \\ & \ddots & & & \\ & & \lambda_r & & \\ & & & 0 & \\ & & & & \ddots \end{bmatrix}, \quad \mathbf{\Lambda}_1 = \begin{bmatrix} \lambda_1 & & & & \\ & \ddots & & & \\ & & \lambda_r & & \\ & & & 0 & \\ & & & & \ddots \end{bmatrix}$$

$(n \times n)$ $(m \times m)$

Note that the numbers of repeated zeros are different!



Basics of SVD: $\mathbf{A}^T\mathbf{A}$ and $\mathbf{A}\mathbf{A}^T$

- Let $\mathbf{v}_i$ be the unit eigenvector of $\mathbf{A}^T\mathbf{A}$ corresponding to a nonzero λ_i
- The corresponding eigenvector for $\mathbf{A}\mathbf{A}^T$: $\mathbf{A}\mathbf{v}_i$, denoted by $\mathbf{u}_i$

$$\|\mathbf{A}\mathbf{v}_i\|^2 = (\mathbf{A}\mathbf{v}_i)^T (\mathbf{A}\mathbf{v}_i) = \lambda_i, \quad i = 1, 2, \dots, r$$

$$\Rightarrow \|\mathbf{A}\mathbf{v}_i\| = \sqrt{\lambda_i}$$

- As a result, the unit eigenvector of $\mathbf{A}\mathbf{A}^T$: $\mathbf{u}_i = \frac{\mathbf{A}\mathbf{v}_i}{\|\mathbf{A}\mathbf{v}_i\|} = \frac{\mathbf{A}\mathbf{v}_i}{\sqrt{\lambda_i}}, \quad i = 1, 2, \dots, r$

$$\mathbf{A}\mathbf{v}_i = \sqrt{\lambda_i} \mathbf{u}_i, \quad i = 1, 2, \dots, r; \quad \mathbf{A}\mathbf{v}_i = 0 \text{ for } i > r$$

$$\Rightarrow \overset{(m \times m)}{\mathbf{A}} \overset{(m \times n)}{\mathbf{Q}_2} = \overset{(m \times n)}{\mathbf{Q}_1} \overset{(n \times n)}{\mathbf{\Sigma}}, \text{ where } \mathbf{\Sigma} = \overset{(m \times n)}{\begin{bmatrix} \sqrt{\lambda_1} & & & \\ & \ddots & & \\ & & \sqrt{\lambda_r} & \\ & & & 0 & & \\ & & & & \ddots & \end{bmatrix}}$$

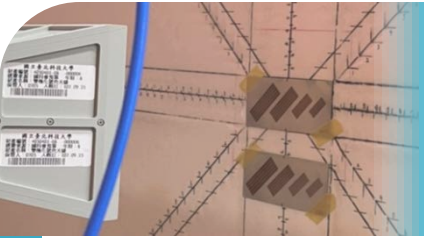
Singular Value Decomposition

- ANY $m \times n$ matrix $\mathbf{A}$ can be factorized into

$$\mathbf{A} = \mathbf{Q}_1 \mathbf{\Sigma} \mathbf{Q}_2^T = (\text{Orthogonal}) \times (\text{Diagonal}) \times (\text{Orthogonal})$$

$$\text{where } \mathbf{\Sigma} = \begin{bmatrix} \sigma_1 & & & & \\ & \ddots & & & \\ & & \sigma_r & & \\ & & & 0 & \\ & & & & \ddots \end{bmatrix} = \begin{bmatrix} \sqrt{\lambda_1} & & & & \\ & \ddots & & & \\ & & \sqrt{\lambda_r} & & \\ & & & 0 & \\ & & & & \ddots \end{bmatrix}$$

- $\sigma_1, \sigma_2, \dots, \sigma_r$ are called singular values
- How do we find $\mathbf{Q}_1$, $\mathbf{\Sigma}$, and $\mathbf{Q}_2$?
 - ◆ Express $\mathbf{A}\mathbf{A}^T$
 - ◆ Similarly, express $\mathbf{A}^T\mathbf{A}$



Characteristics of SVD

- The factorization $\mathbf{A} = \mathbf{Q}_1 \mathbf{\Sigma} \mathbf{Q}_2^T$ is not unique
- Usually, singular values are arranged according to their magnitude
- For positive (semi)definite matrices, $\mathbf{A} = \mathbf{Q}_1 \mathbf{\Sigma} \mathbf{Q}_2^T = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^T$
- For complex matrices, $\mathbf{\Sigma}$ remains real, whereas $\mathbf{Q}_1$ and $\mathbf{Q}_2$ are unitary matrices
- Geometric meaning of SVD: $\mathbf{A}\mathbf{x} = \mathbf{Q}_1 \mathbf{\Sigma} \mathbf{Q}_2^T \mathbf{x}$

$\mathbf{x}$	$\mathbf{Q}_2^T \mathbf{x}$	$\mathbf{\Sigma} \mathbf{Q}_2^T \mathbf{x}$	$\mathbf{Q}_1 \mathbf{\Sigma} \mathbf{Q}_2^T \mathbf{x}$

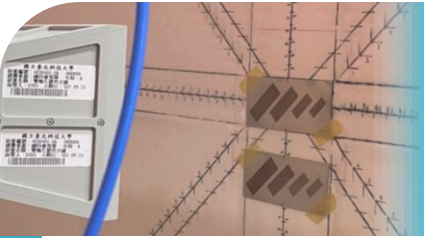
Examples

- Example 1: $\mathbf{A}$ is diagonal

$$\mathbf{A} = \begin{bmatrix} 2 & 0 \\ 0 & -3 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

- Example 2: $\mathbf{A}$ has only one column

$$\mathbf{A} = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} & -\frac{1}{3} \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix} [1]$$



Examples

- Example 3: $\mathbf{A}$ has only one row

$$\mathbf{A} = \begin{bmatrix} -1 & 2 & 2 \end{bmatrix}$$

- Example 4: $\mathbf{A}$ is already orthogonal

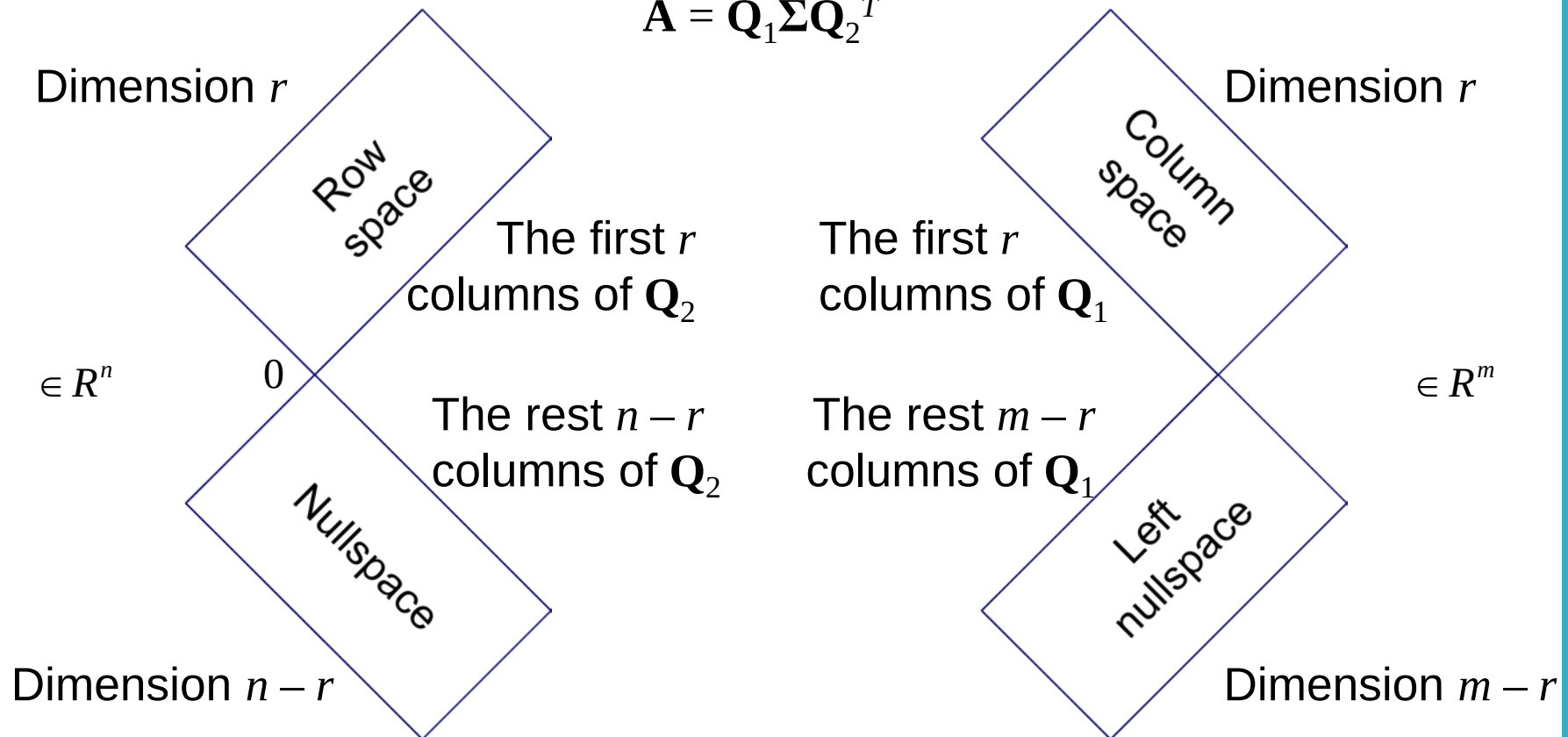
$$\mathbf{A} = \mathbf{Q}_1 \mathbf{\Sigma} \mathbf{Q}_2^T$$

- Example 5: $\mathbf{A}$ is an incidence matrix

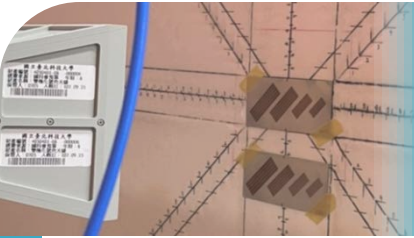
$$\mathbf{A} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{-2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{-1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix}$$

Bases for 4 Fundamental Subspaces

$$\mathbf{A} = \mathbf{Q}_1 \mathbf{\Sigma} \mathbf{Q}_2^T$$



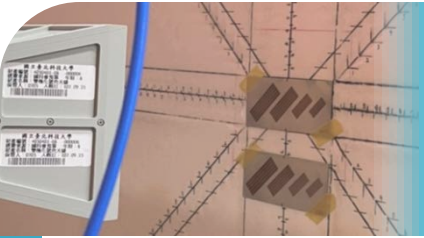
SVD finds the basis for the 4 subspaces!



An Example

$$\mathbf{A} = \begin{bmatrix} 1/2 & 1/2 & 1/2 & 1/2 \\ 1/2 & -1/2 & -1/2 & 1/2 \\ 1/2 & -1/2 & 1/2 & -1/2 \\ 1/2 & 1/2 & -1/2 & -1/2 \end{bmatrix} \begin{bmatrix} 12 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 2/3 & 2/3 & 1/3 \\ -2/3 & 1/3 & 2/3 \\ 1/3 & -2/3 & 2/3 \end{bmatrix}$$

1. What is the rank of $\mathbf{A}$?
2. What are the eigenvalues for $\mathbf{A}^T\mathbf{A}$?
3. What are the eigenvalues for $\mathbf{A}\mathbf{A}^T$?
4. What is the basis for the column space of $\mathbf{A}$?
5. Based on the previous query, apply Gram-Schmidt to make the basis orthonormal
6. What is the basis for the nullspace of $\mathbf{A}$?
7. What is the basis for the row space of $\mathbf{A}$?
8. What is the basis for the left nullspace of $\mathbf{A}$?



Agenda

1

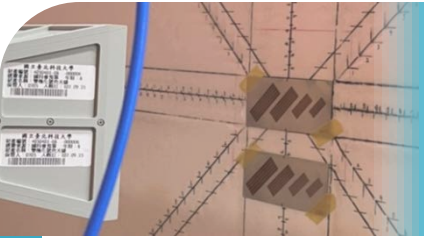
Concept

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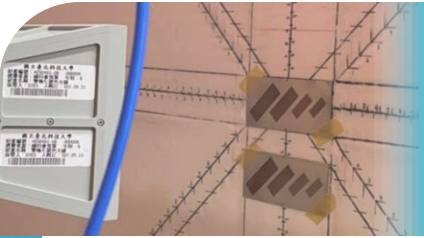


Norm of a Matrix (from Ch. 3)

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & & a_{2n} \\ \vdots & & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

- A matrix also can be defined with a norm, which measures the size
- σ_i : the i^{th} singular value

$\ \mathbf{A}\ _2$ norm	$\ \mathbf{A}\ _2 = \sigma_1$
$\ \mathbf{A}\ _F$ norm (Frobenius norm)	$\ \mathbf{A}\ _F = \sqrt{ a_{11} ^2 + a_{12} ^2 + \cdots + a_{mn} ^2}$
$\ \mathbf{A}\ _N$ norm (Nuclear norm)	$\ \mathbf{A}\ _N = \sigma_1 + \sigma_2 + \cdots + \sigma_r$



1. Image Processing

- Satellite takes a picture containing $1,000 \times 1,000$ pixels represented by matrix $\mathbf{A}$; each pixels with a color number $\rightarrow$ to send 1,000,000 numbers
- SVD of Image $\mathbf{A}$:

$$\mathbf{A} = \mathbf{Q}_1 \mathbf{\Sigma} \mathbf{Q}_2^T = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \dots + \sigma_r \mathbf{u}_r \mathbf{v}_r^T$$

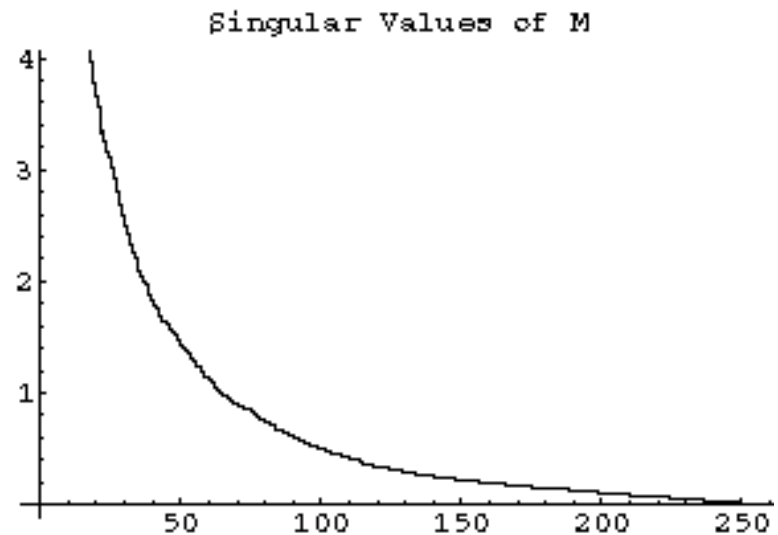
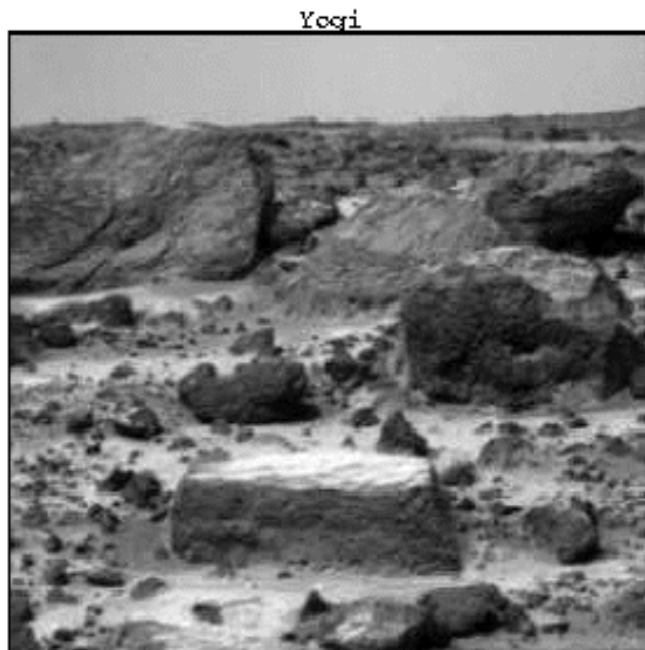
- We may keep only the first few terms with larger singular values, say 60 of them:

$$\mathbf{A} = \mathbf{Q}_1 \mathbf{\Sigma} \mathbf{Q}_2^T = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \dots + \sigma_{60} \mathbf{u}_{60} \mathbf{v}_{60}^T$$

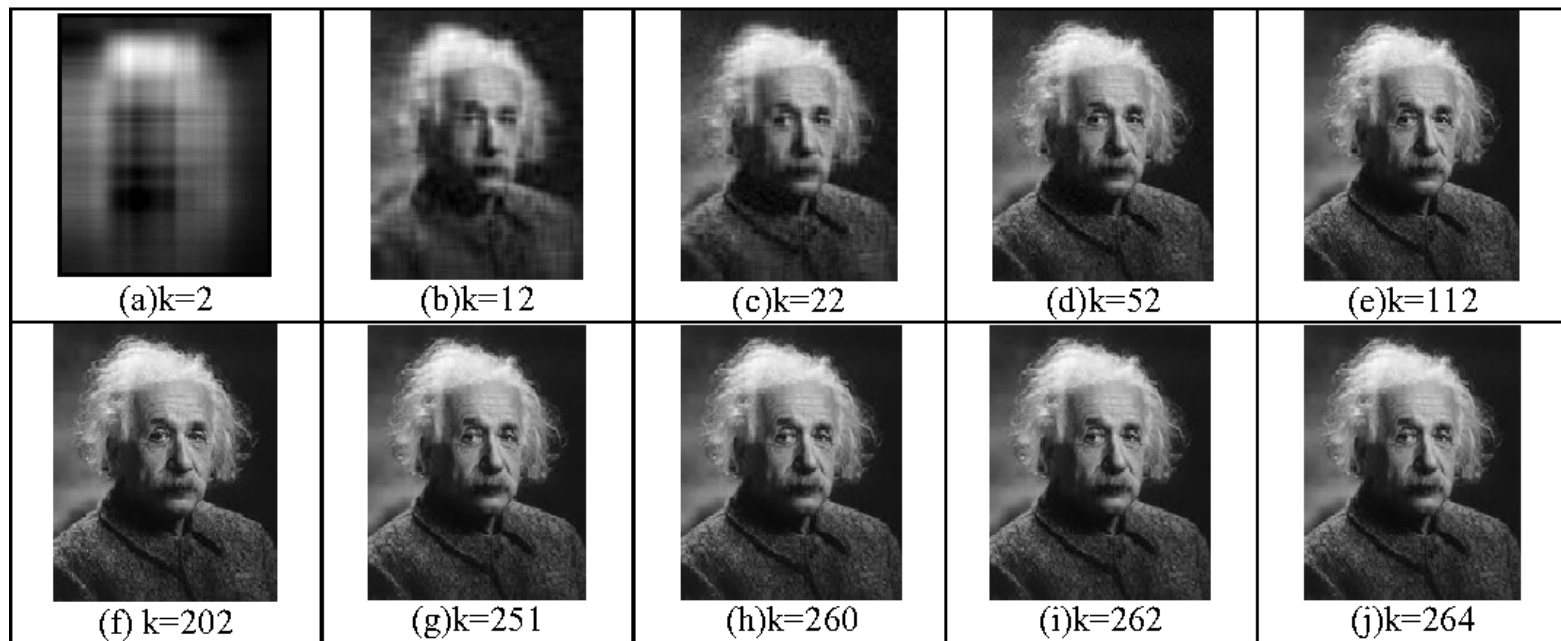
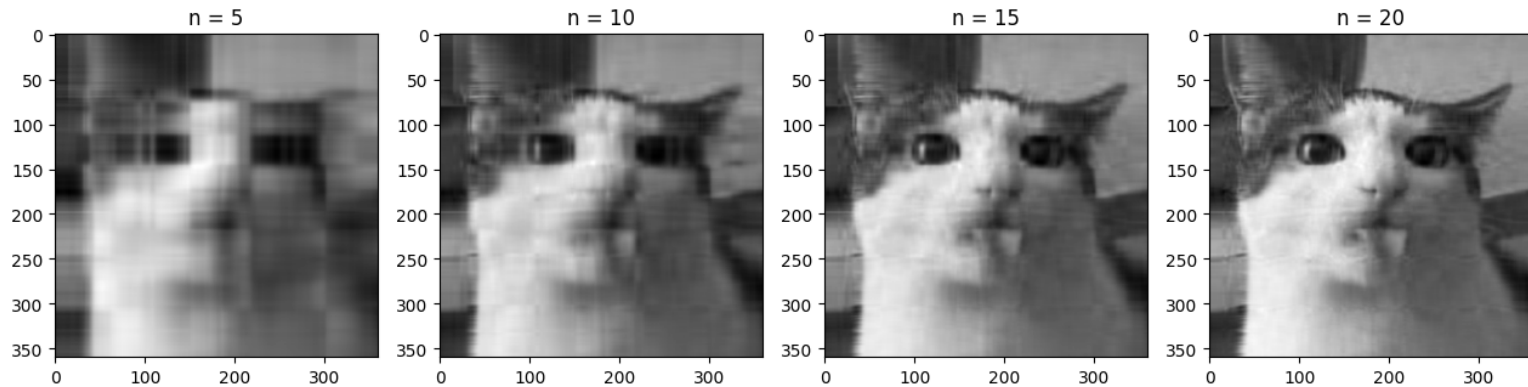
and throw away 940 terms and only send $60 + 60 \times 2000$ back

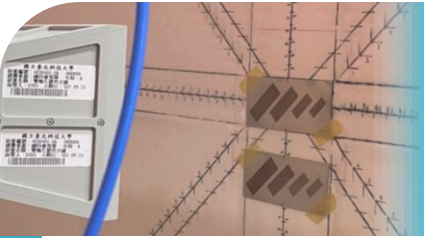
1. Image Processing

- The image gets more and more lucid as more singular-value terms are added
- Example: Martian image of a rock called "Yogi" by the Sojourner rover: 256×264



1. Image Processing





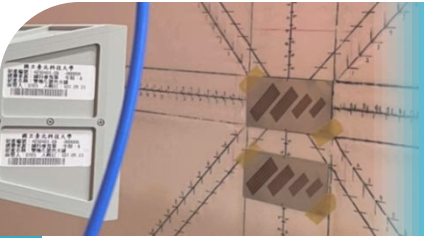
2. Polar Decomposition

- Robot arm: rotate and stretch-out/draw-back
- Orthopedics: Rotation and Stretching/Compression of a certain part of your body
- Material deformation expressed by a matrix $\mathbf{A}$
- Every real square matrix can be factored into

$$\mathbf{A} = \mathbf{Q}\mathbf{S}$$

where $\mathbf{Q}$ is orthogonal and $\mathbf{S}$ is symmetric positive (semi)definite

- Proof: $\mathbf{A} = \mathbf{Q}_1\mathbf{\Sigma}\mathbf{Q}_2^T = (\mathbf{Q}_1\mathbf{Q}_2^T)(\mathbf{Q}_2\mathbf{\Sigma}\mathbf{Q}_2^T) \rightarrow \mathbf{Q} = \mathbf{Q}_1\mathbf{Q}_2^T; \mathbf{S} = \mathbf{Q}_2\mathbf{\Sigma}\mathbf{Q}_2^T$

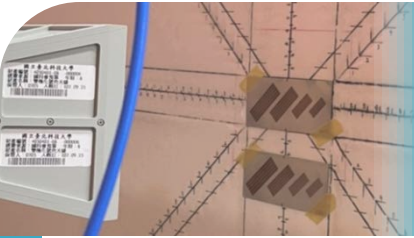


2. Polar Decomposition

- Meaning of $\mathbf{A} = \mathbf{Q}\mathbf{S}$ ($= (\mathbf{Q}_1\mathbf{Q}_2^T)(\mathbf{Q}_2\mathbf{\Sigma}\mathbf{Q}_2^T)$)
 - ◆ $\mathbf{Q}$: orthogonal; rotation or reflection
 - ◆ $\mathbf{S}$: symmetric and positive (semi)definite; eigenvalues $\sigma_1, \sigma_2, \dots, \sigma_r \rightarrow$ stretch or compress through the directions of columns in $\mathbf{Q}_2$
- Example:

$$\mathbf{A} = \begin{bmatrix} 1 & -2 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$$

$$\mathbf{A} = \begin{bmatrix} 1 & -2 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$



3. Pseudoinverse

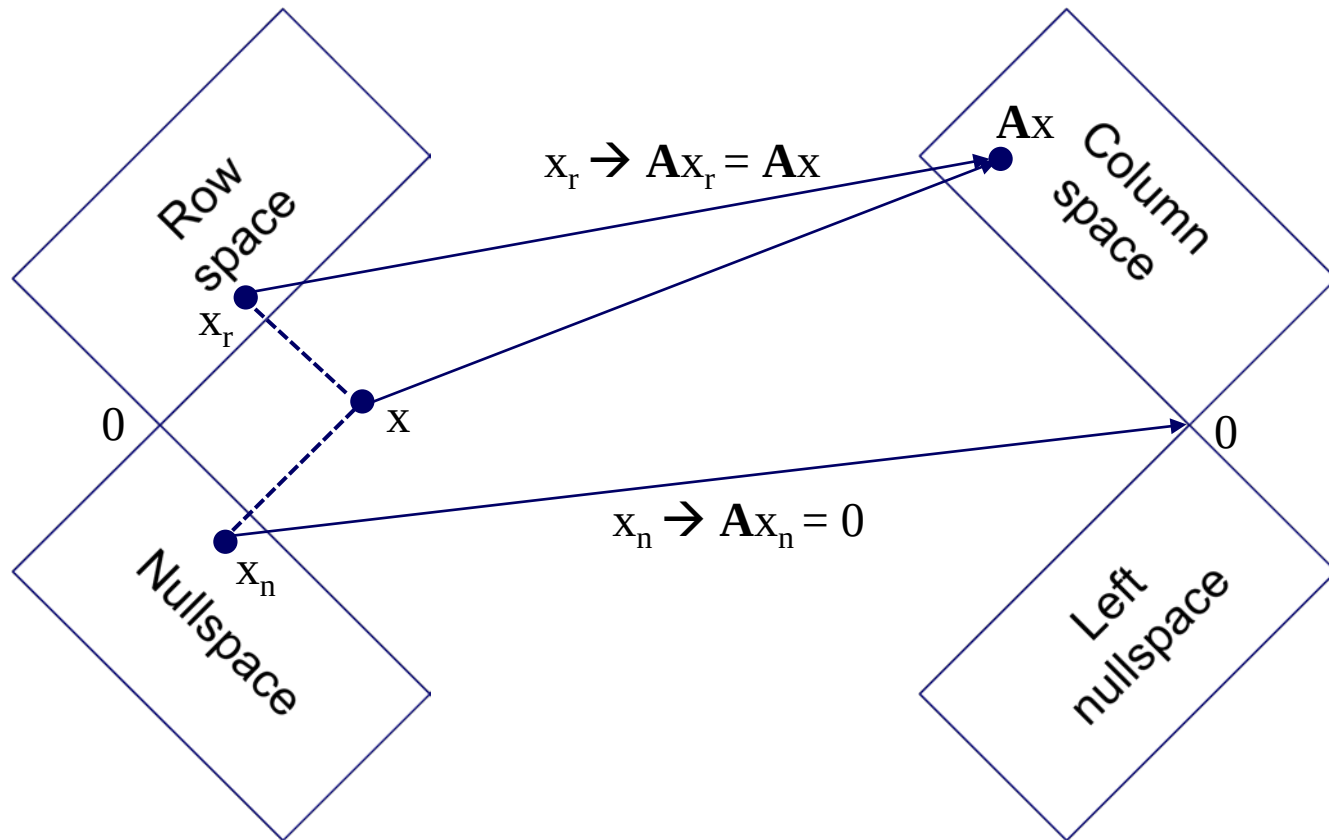
$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

- Rows of $\mathbf{A}$ are dependent $\rightarrow$ Likely no solution $\rightarrow$ Find $\mathbf{A}^T \mathbf{A} \hat{\mathbf{x}} = \mathbf{A}^T \mathbf{b}$
- Columns of $\mathbf{A}$ are dependent $\rightarrow \mathbf{A}^T \mathbf{A}$ is not invertible!
- How do we deal with this situation?

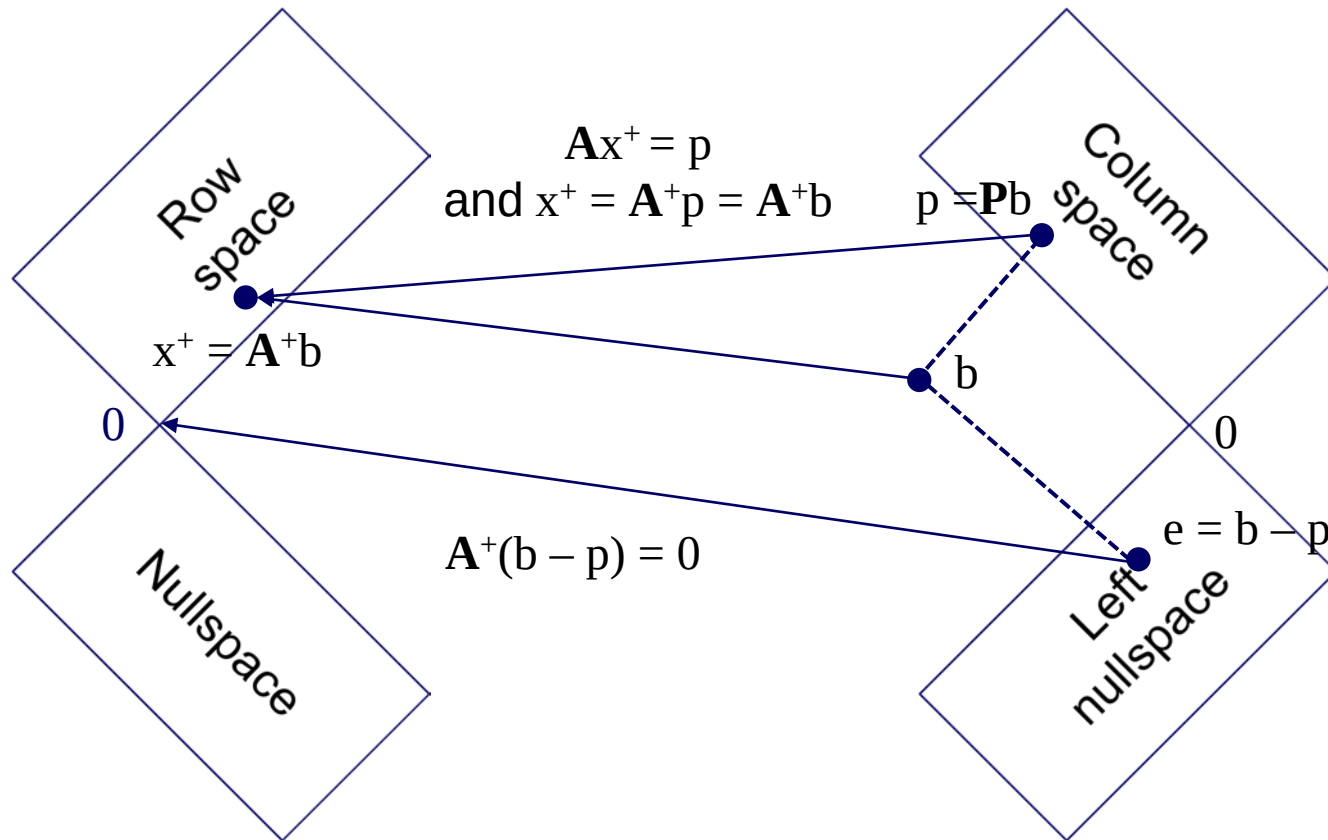


- Optimal solution: the minimum length solution of $\mathbf{A}^T \mathbf{A} \hat{\mathbf{x}} = \mathbf{A}^T \mathbf{b}$
- This minimum length solution is called $\mathbf{x}^+$
- $\mathbf{x}^+$ summarizes the solution for solvable $\mathbf{A}\mathbf{x} = \mathbf{b}$, unsolvable $\mathbf{A}\hat{\mathbf{x}} = \mathbf{b}$, and non-invertible $\mathbf{A}^T \mathbf{A}$ in $\mathbf{A}^T \mathbf{A} \hat{\mathbf{x}} = \mathbf{A}^T \mathbf{b}$

3. Pseudoinverse



3. Pseudoinverse



3. Pseudoinverse

- Characteristics of $\mathbf{A}^+$:
 - ◆ $(\mathbf{A}^+)^+ = \mathbf{A}$
 - ◆ When $\mathbf{A}$ is invertible, $\mathbf{A}^+ = \mathbf{A}^{-1}$
- Least square solution $\mathbf{A}\hat{\mathbf{x}} = \mathbf{p} \rightarrow$ Split into row space component and nullspace component:

$$\hat{\mathbf{x}} = \hat{\mathbf{x}}_r + \hat{\mathbf{x}}_n \Rightarrow \|\hat{\mathbf{x}}\|^2 = \|\hat{\mathbf{x}}_r\|^2 + \|\hat{\mathbf{x}}_n\|^2$$



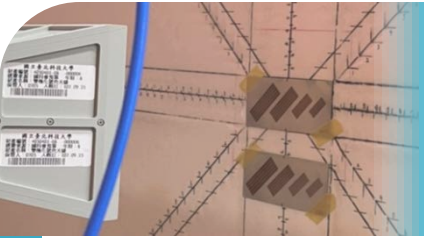
$$\mathbf{A}\hat{\mathbf{x}}_r = \mathbf{p}$$

$$\mathbf{A}\hat{\mathbf{x}}_n = 0$$



$\hat{\mathbf{x}}$ is shortest when $\|\hat{\mathbf{x}}_n\| = 0$

All $\hat{\mathbf{x}}$ has the same row space component: $\mathbf{x}^+ = \hat{\mathbf{x}}_r$



3. Pseudoinverse

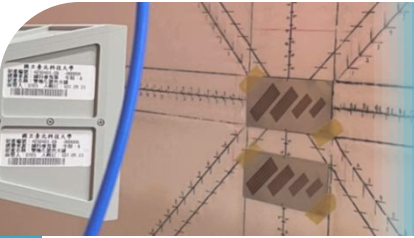
- Example: $\mathbf{A}$ is diagonal, with dependent rows and dependent columns:

$$\mathbf{A}\hat{\mathbf{x}} = \mathbf{b} \Rightarrow \begin{bmatrix} \sigma_1 & 0 & 0 & 0 \\ 0 & \sigma_2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \\ \hat{x}_3 \\ \hat{x}_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

- Solving $\mathbf{A}\hat{\mathbf{x}} = \mathbf{p}$ instead:

$$\mathbf{A}\hat{\mathbf{x}} = \mathbf{p} \Rightarrow \begin{bmatrix} \sigma_1 & 0 & 0 & 0 \\ 0 & \sigma_2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \\ \hat{x}_3 \\ \hat{x}_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ 0 \end{bmatrix} \Rightarrow \hat{x}_1 = \frac{b_1}{\sigma_1}, \hat{x}_2 = \frac{b_2}{\sigma_2}$$

- Columns are dependent, so $\hat{x}_3$ and $\hat{x}_4$ can be chosen arbitrary
- However, the optimal solution is $\hat{x}_3 = \hat{x}_4 = 0$ because of minimum norm for $\mathbf{x}$



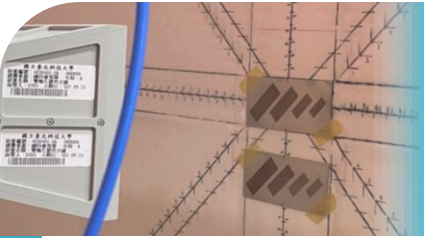
3. Pseudoinverse

- Such optimal solution (x^+) can be expressed as:

$$x^+ = \begin{bmatrix} \frac{b_1}{\sigma_1} \\ \frac{b_2}{\sigma_2} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sigma_1} & 0 & 0 \\ 0 & \frac{1}{\sigma_2} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

- This expression displays the matrix that produces x^+ from b
- Such a matrix is the pseudoinverse A^+

$$Ax = b \quad \Rightarrow \quad x^+ = A^+b$$



3. Pseudoinverse

- Example: Solve $-x_1 + 2x_2 + 2x_3 = 18$

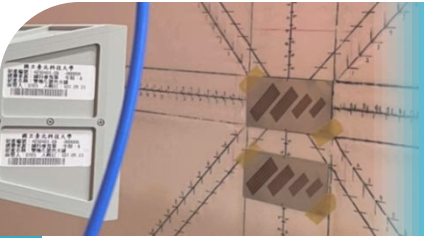
$$\rightarrow x^+ = [-2 \ 4 \ 4]^T$$

$\rightarrow$ How about $x = [-2 \ 5 \ 3]^T$, $[-6 \ 3 \ 3]^T$, and $[-2 \ 7 \ 1]^T$?

- Example: In Ch. 2

$$\begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 5 \end{bmatrix} \rightarrow x = \begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

$$\rightarrow x^+ = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$



3. Pseudoinverse

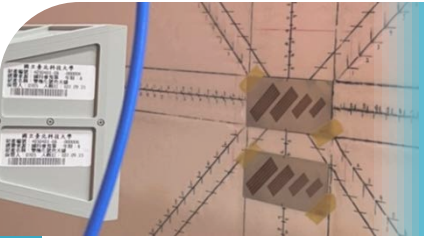
Formula of A^+

$$\mathbf{A} = \mathbf{Q}_1 \mathbf{\Sigma} \mathbf{Q}_2^T \rightarrow \mathbf{A}^+ = \mathbf{Q}_2 \mathbf{\Sigma}^+ \mathbf{Q}_1^T, \text{ where}$$

$(n \times m)$

$$\mathbf{\Sigma}^+ = \begin{bmatrix} \frac{1}{\sigma_1} & & & & \\ & \ddots & & & \\ & & \frac{1}{\sigma_r} & & \\ & & & 0 & \\ & & & & \ddots \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{\lambda_1}} & & & & \\ & \ddots & & & \\ & & \frac{1}{\sqrt{\lambda_r}} & & \\ & & & 0 & \\ & & & & \ddots \end{bmatrix}$$

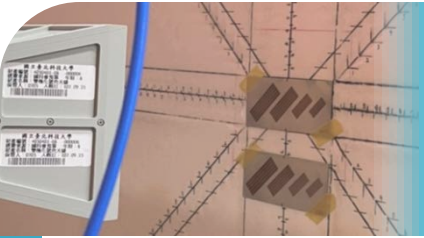
$(n \times m)$



3. Pseudoinverse

- Example: Find the optimal solution for $-x_1 + 2x_2 + 2x_3 = 18$
 - ◆ Step 1: Formulate $\mathbf{A}\mathbf{x} = \mathbf{b}$
 - ◆ Step 2: Find the SVD for $\mathbf{A}$
 - ◆ Step 3: Find $\mathbf{A}^+$
 - ◆ Step 4: $\mathbf{x}^+ = \mathbf{A}^+\mathbf{b}$

$$\mathbf{A} = \begin{bmatrix} -1 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 \end{bmatrix} \begin{bmatrix} 3 & 0 & 0 \end{bmatrix} \begin{bmatrix} -\frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} & -\frac{1}{3} \end{bmatrix}$$



3. Pseudoinverse

Proof of $\mathbf{A}^+$ formula:

- Multiplication by the orthogonal matrix $\mathbf{Q}_1^T$ leaves lengths unchanged:

$$\|\mathbf{Ax} - \mathbf{b}\| = \|\mathbf{Q}_1 \boldsymbol{\Sigma} \mathbf{Q}_2^T \mathbf{x} - \mathbf{b}\| = \|\mathbf{Q}_1^T (\mathbf{Q}_1 \boldsymbol{\Sigma} \mathbf{Q}_2^T \mathbf{x} - \mathbf{b})\| = \|\boldsymbol{\Sigma} \mathbf{Q}_2^T \mathbf{x} - \mathbf{Q}_1^T \mathbf{b}\|$$

- Introduce the new unknown $\mathbf{y} = \mathbf{Q}_2^T \mathbf{x} = \mathbf{Q}_2^{-1} \mathbf{x}$, which has the same length as $\mathbf{x}$
- Then, minimizing $\|\mathbf{Ax} - \mathbf{b}\|$ is the same as minimizing $\|\boldsymbol{\Sigma} \mathbf{y} - \mathbf{Q}_1^T \mathbf{b}\|$
- Now $\boldsymbol{\Sigma}$ is diagonal and we know the best $\mathbf{y}^+$
- It is $\mathbf{y}^+ = \boldsymbol{\Sigma}^+ \mathbf{Q}_1^T \mathbf{b}$, so the best $\mathbf{x}^+ = \mathbf{Q}_2 \mathbf{y}^+ = \mathbf{Q}_2 \boldsymbol{\Sigma}^+ \mathbf{Q}_1^T \mathbf{b} = \mathbf{A}^+ \mathbf{b}$